

S1: Buenos Aires

```
interface GigabitEthernet0/0.10
description mS1
no shutdown
encapsulation dot1Q 10
ip address 172.16.10.1 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.20
description sS1
no shutdown
encapsulation dot1Q 20
ip address 172.16.10.33 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.30
description wS1
no shutdown
encapsulation dot1Q 30
ip address 172.16.10.65 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.40
description svS1
no shutdown
encapsulation dot1Q 40
ip address 172.16.10.97 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.50
description gS1
no shutdown
encapsulation dot1Q 50
ip address 172.16.10.129 255.255.255.224
exit
```

```
router rip
version 1
network 172.16.10.0
network 5.0.0.0
network 6.0.0.0
exit
```

```
ip dhcp excluded-address 172.16.10.1 172.16.10.10
```

```
ip dhcp pool mS1
network 172.16.10.0 255.255.255.224
default-router 172.16.10.1
dns-server 5.5.5.5
exit
```

```
ip dhcp pool sS1
network 172.16.10.32 255.255.255.224
default-router 172.16.10.33
dns-server 5.5.5.5
exit
```

```
ip dhcp pool wS1
network 172.16.10.64 255.255.255.224
default-router 172.16.10.65
dns-server 5.5.5.5
exit
```

```
ip dhcp pool gS1
network 172.16.10.128 255.255.255.224
default-router 172.16.10.129
dns-server 5.5.5.5
exit
```

```
copy running-config startup-config
```

S2: Córdoba

```
interface GigabitEthernet0/0.10
description mS1
no shutdown
encapsulation dot1Q 10
ip address 192.168.100.1 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.20
description sS1
no shutdown
encapsulation dot1Q 20
ip address 192.168.100.33 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.30
description wS1
no shutdown
encapsulation dot1Q 30
ip address 192.168.100.65 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.40
description svS1
no shutdown
encapsulation dot1Q 40
ip address 192.168.100.97 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.50
description gS1
no shutdown
encapsulation dot1Q 50
ip address 192.168.100.129 255.255.255.224
exit
```

```
router rip
version 1
network 192.168.100.0
network 5.0.0.0
network 6.0.0.0
exit
```

```
ip dhcp pool mS1
network 192.168.100.0 255.255.255.224
default-router 172.16.10.1
```

```
dns-server 5.5.5.5
exit
```

```
ip dhcp pool sS1
network 192.168.100.32 255.255.255.224
default-router 192.168.100.33
dns-server 5.5.5.5
exit
```

```
ip dhcp pool wS1
network 192.168.100.64 255.255.255.224
default-router 192.168.100.65
dns-server 5.5.5.5
exit
```

```
ip dhcp pool gS1
network 192.168.100.128 255.255.255.224
default-router 192.168.100.129
dns-server 5.5.5.5
exit
```

```
copy running-config startup-config
```

S3: Tierra del Fuego

```
interface GigabitEthernet0/0.10
description ManagementS3
no shutdown
encapsulation dot1Q 10
ip address 203.0.115.1 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.20
description StaffS3
no shutdown
encapsulation dot1Q 20
ip address 203.0.115.33 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.30
description WifiS3
no shutdown
encapsulation dot1Q 30
ip address 203.0.115.65 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.40
```

```
description ServersS3
no shutdown
encapsulation dot1Q 40
ip address 203.0.115.97 255.255.255.224
exit
```

```
interface GigabitEthernet0/0.50
description GuestS3
no shutdown
encapsulation dot1Q 50
ip address 203.0.115.129 255.255.255.224
exit
```

```
ip dhcp pool ManagementS3
network 203.0.115.0 255.255.255.224
default-router 203.0.115.1
dns-server 5.5.5.5
exit
```

```
ip dhcp pool StaffS3
network 203.0.115.32 255.255.255.224
default-router 203.0.115.33
dns-server 5.5.5.5
exit
```

```
ip dhcp pool WifiS3
network 203.0.115.64 255.255.255.224
default-router 203.0.115.65
dns-server 5.5.5.5
exit
```

```
ip dhcp pool GuestS3
network 203.0.115.128 255.255.255.224
default-router 203.0.115.129
dns-server 5.5.5.5
exit
```

```
router rip
version 1
network 192.168.100.0
network 11.0.0.0
network 12.0.0.0
exit
```

```
copy running-config startup-config
```

TUNELES

S1

```
interface Serial0/3/0
  ip address 20.0.0.1 255.255.255.224
  no shutdown

! Tunnel a S2
interface Tunnel10
  ip address 10.10.10.1 255.255.255.224
  tunnel source Serial0/3/0
  tunnel destination 20.0.0.2
  no shutdown

! Tunnel a S3
interface Tunnel11
  ip address 10.10.10.33 255.255.255.224
  tunnel source Serial0/3/0
  tunnel destination 20.0.0.6
  no shutdown
```

S2

```
interface Serial0/3/0
  ip address 20.0.0.2 255.255.255.224
  no shutdown

! Tunnel a S1
interface Tunnel10
  ip address 10.10.10.2 255.255.255.224
  tunnel source Serial0/3/0
  tunnel destination 20.0.0.1
  no shutdown

! Tunnel a S3
interface Tunnel11
  ip address 10.10.10.65 255.255.255.224
```

```
tunnel source Serial0/3/0
tunnel destination 20.0.0.6
no shutdown
```

```
ip route 10.10.10.0 255.255.255.224 20.0.0.1    ! hacia S1
ip route 10.10.10.32 255.255.255.224 20.0.0.6    ! hacia S3 (si
prefieres via S3 directamente)
```

S3

```
interface Serial0/3/0
ip address 20.0.0.6 255.255.255.224
no shutdown
```

```
! Tunnel a S1
interface Tunnel10
ip address 10.10.10.34 255.255.255.224
tunnel source Serial0/3/0
tunnel destination 20.0.0.1
no shutdown
```

```
!
! Tunnel a S2
interface Tunnel11
ip address 10.10.10.66 255.255.255.224
tunnel source Serial0/3/0
tunnel destination 20.0.0.2
no shutdown
```

ACL

```
R1
ip access-list extended ACL-S1
permit ip 172.16.10.0 0.0.0.31 172.16.10.96 0.0.0.31
permit ip 172.16.10.32 0.0.0.31 172.16.10.96 0.0.0.31
permit ip 172.16.10.64 0.0.0.31 172.16.10.96 0.0.0.31
permit ip 172.16.10.128 0.0.0.31 172.16.10.96 0.0.0.31
permit udp 172.16.10.0 0.0.0.31 host 6.0.0.1 eq domain
permit udp 172.16.10.32 0.0.0.31 host 6.0.0.1 eq domain
```

```
permit udp 172.16.10.64 0.0.0.31 host 6.0.0.1 eq domain
permit udp 172.16.10.128 0.0.0.31 host 6.0.0.1 eq domain
permit tcp 172.16.10.0 0.0.0.31 host 6.0.0.1 eq 80
permit tcp 172.16.10.0 0.0.0.31 host 6.0.0.1 eq 443
deny tcp any host 172.16.10.101 eq 80
deny tcp any host 172.16.10.101 eq 443
permit tcp 172.16.10.0 0.0.0.31 any eq 22

remark
deny ip any any
```

R2

```
ip access-list extended ACL-S2
permit ip 192.168.100.80 0.0.0.15 192.168.100.96 0.0.0.7
permit ip 192.168.100.32 0.0.0.31 192.168.100.96 0.0.0.7
permit ip 192.168.100.0 0.0.0.31 192.168.100.96 0.0.0.7
permit ip 192.168.100.64 0.0.0.15 192.168.100.96 0.0.0.7
permit udp 192.168.100.80 0.0.0.15 host 6.0.0.1 eq domain
permit udp 192.168.100.32 0.0.0.31 host 6.0.0.1 eq domain
permit udp 192.168.100.0 0.0.0.31 host 6.0.0.1 eq domain
permit udp 192.168.100.64 0.0.0.15 host 6.0.0.1 eq domain
permit tcp 192.168.100.80 0.0.0.15 host 6.0.0.1 eq 80
permit tcp 192.168.100.80 0.0.0.15 host 6.0.0.1 eq 443
deny tcp any host 192.168.100.81 eq 80
deny tcp any host 192.168.100.81 eq 443
permit tcp 192.168.100.80 0.0.0.15 any eq 22
deny ip any any
```

R3

```
ip access-list extended ACL-S3
permit ip 203.0.115.0 0.0.0.31 203.0.115.96 0.0.0.31
permit ip 203.0.115.32 0.0.0.31 203.0.115.96 0.0.0.31
```



```
permit ip 203.0.115.64 0.0.0.31 203.0.115.96 0.0.0.31
permit ip 203.0.115.128 0.0.0.31 203.0.115.96 0.0.0.31
permit udp 203.0.115.0 0.0.0.31 host 6.0.0.1 eq domain
permit udp 203.0.115.32 0.0.0.31 host 6.0.0.1 eq domain
permit udp 203.0.115.64 0.0.0.31 host 6.0.0.1 eq domain
permit udp 203.0.115.128 0.0.0.31 host 6.0.0.1 eq domain
permit tcp 203.0.115.0 0.0.0.31 host 6.0.0.1 eq 80
permit tcp 203.0.115.0 0.0.0.31 host 6.0.0.1 eq 443
deny tcp any host 203.0.115.2 eq 80
deny tcp any host 203.0.115.2 eq 443
permit tcp 203.0.115.0 0.0.0.31 any eq 22
deny ip any any
```

```
config de router de redundancia
enable
configure terminal
hostname R-RED
no ip domain-lookup
```

```
interface GigabitEthernet0/0
ip address 6.0.0.2 255.255.255.252
no shutdown
```

```
interface GigabitEthernet0/1
ip address 6.0.0.6 255.255.255.252
no shutdown
```

```
interface GigabitEthernet0/2
ip address 6.0.0.10 255.255.255.252
no shutdown
```

```
interface Tunnel0
ip address 10.20.0.2 255.255.255.252
tunnel source GigabitEthernet0/0
tunnel destination 6.0.0.1
no shutdown
```

```
interface Tunnel1
```

```
ip address 10.20.0.6 255.255.255.252
tunnel source GigabitEthernet0/1
tunnel destination 6.0.0.5
no shutdown
```

```
interface Tunnel2
ip address 10.20.0.10 255.255.255.252
tunnel source GigabitEthernet0/2
tunnel destination 6.0.0.9
no shutdown
```

```
router rip
version 2
no auto-summary
network 6.0.0.0
network 10.0.0.0
network 172.16.0.0
```

```
ip name-server 172.16.0.10
```

```
end
write memory
```

VPN

```
S1
enable
configure terminal
```

```
interface Tunnel5
ip address 10.20.0.1 255.255.255.252
tunnel source Serial0/3/0
tunnel destination 6.0.0.2
no shutdown
```

```
router rip
version 2
no auto-summary
network 10.0.0.0
network 6.0.0.0
```

```
end
write memory
```

```
S2
enable
configure terminal
```

```
interface Tunnel6
ip address 10.20.0.5 255.255.255.252
tunnel source Serial0/3/1
tunnel destination 6.0.0.6
no shutdown
```

```
router rip
version 2
no auto-summary
network 10.0.0.0
network 6.0.0.0
```

```
S3
interface Tunnel7
ip address 10.20.0.9 255.255.255.252
tunnel source Serial0/3/0
tunnel destination 6.0.0.10
no shutdown
```

```
router rip
version 2
no auto-summary
network 10.0.0.0
network 6.0.0.0
```

```
end
write memory
```